Robust Hippocampal-Striatal Recruitment During Auditory Processing: A Non-Selective Response to Novelty

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***Abstract*—The brain's ability to detect and adapt to novel stimuli is fundamental for survival. While local neural responses such as the P300 are well-documented, the dynamic network interactions underlying these processes—specifically the directional communication between the Hippocampus and Striatum—remain poorly understood. This study addresses whether the neural processing of novelty is mediated by a transient reorganization of information flow in these circuits.**

We identified a robust, time-locked increase in Hippocampal-Striatal (HC-dS) connectivity occurring 330–730 ms post-stimulus. However, contrary to the hypothesis that this circuit acts as a selective novelty filter, this modulation was observed equally for both Standard and Target stimuli, with no statistically significant difference between conditions ($p > 0.05$). This finding challenges the view of the hippocampal-striatal loop as a purely novelty-gated filter in passive settings, suggesting instead that it is recruited for general temporal integration.

To arrive at these conclusions, we analyzed Local Field Potential (LFP) recordings from the Hippocampus, Dorsal Striatum, and mPFC during a passive auditory oddball task. We employed time-resolved Phase-Amplitude Coupling (PAC) and Granger Causality (GC) analyses to quantify the dynamic evolution of directional information flow across 200 ms sliding windows.

I. Introduction

A. The Significance of Novelty and Surprise

Survival and adaptive behavior critically depend on the brain's ability to anticipate future events and rapidly detect violations of those expectations. When incoming sensory information deviates from internal models, the resulting experience of "surprise" signals that existing knowledge may be incomplete. This process is not merely a passive reaction but a fundamental driver of cognitive flexibility; by signaling a mismatch between prediction and reality, surprise facilitates the revision of internal models and triggers essential learning mechanisms [1], [2]. Specifically, theoretical frameworks distinguish between "puzzlement" surprise, which signals a need for model update, and "enlightenment" surprise, reflecting successful integration of information [3].

Impairments in novelty and surprise processing have been increasingly implicated in psychiatric disorders. For instance, predictive coding frameworks suggest that conditions such as schizophrenia and major depression arise from dysfunctional inference and aberrant weighting of prediction errors, making the study of these mechanisms crucial for understanding mental health [4], [5].

B. Neural Markers: MMN and P300

Traditionally, the neural correlates of novelty have been studied using Event-Related Potentials (ERPs). The Mismatch Negativity (MMN) is a well-established early component reflecting the automatic detection of sensory violations, often attributed to a pre-attentive comparison between sensory input and memory traces [6], [7]. A later component, the P300 (specifically P3a and P3b), is associated with attention allocation and the updating of internal context models [8]. While these markers have provided valuable insights, they primarily describe local or averaged neural responses and do not fully capture the dynamic, network-level communication required for complex model updating [9].

C. The Oddball Paradigm and LFP Recordings

To investigate these mechanisms, the auditory oddball task serves as a powerful experimental paradigm. In this task, a sequence of frequent "standard" stimuli is interspersed with rare "target" (deviant) stimuli, effectively inducing prediction errors. While scalp EEG captures global cortical responses, it lacks the spatial resolution to resolve activity in deep subcortical structures. Local Field Potential (LFP) recordings offer a distinct advantage by allowing direct measurement of neural activity from specific brain regions. This high spatial and temporal resolution is essential for dissecting the precise timing and directionality of interactions between distinct neural circuits [10].

D. The Hippocampus-Striatum-Prefrontal Circuit

Novelty detection is not a localized process but emerges from the coordinated activity of a distributed network. The **Hippocampus (HC)** is hypothesized to act as a comparator, detecting discrepancies between expected and observed sensory inputs and regulating the entry of information into long-term memory [11], [12]. The **Dorsal Striatum (dS)** plays a central role in learning from prediction errors and updating action-outcome associations, often modulated by dopaminergic signals [13]. Meanwhile, the **Medial Prefrontal**

Cortex (mPFC) is involved in higher-order cognitive control and the integration of information over time [14], [15].

Crucially, these regions do not operate in isolation. Theoretical models suggest that the interaction between them is vital for adaptive behavior. For instance, the hippocampus is thought to gate information flow to the striatum and cortex, thereby facilitating the encoding of novel events [11]. The coordination of this circuitry is essential for translating sensory surprise into the behavioral adjustments required for learning.

E. Gap Identification

Despite the established roles of these individual regions, a significant conceptual gap remains regarding the *temporal dynamics of information flow* within this network. Most prior studies have focused on static connectivity measures or isolated regional responses, leaving it unclear how prediction error signals propagate across the HC-Striatum-mPFC circuit over time [10]. Specifically, we lack a clear understanding of how the directionality and strength of these interactions modulate dynamically during the brief latency of cognitive processing (e.g., the P300 time window) following a surprising event.

F. Novel Hypothesis

To address the identified gap regarding temporal network dynamics, this study investigates functional connectivity within the hippocampal-striatal-prefrontal circuit during an auditory oddball task. **Our central research question asks whether the neural processing of novelty is mediated by a transient reorganization of directional information flow, rather than localized activation alone.**

Based on the proposed role of the hippocampus as a comparator, we formulate the following testable hypothesis: the presentation of a *target (novel)* stimulus, compared to a *standard* stimulus, will trigger a significant increase in directional coupling from the **Hippocampus (HC)** to the **Dorsal Striatum (dS)** and **Medial Prefrontal Cortex (mPFC)**.

Specifically, we predict that this interaction will be manifested as enhanced **Phase-Amplitude Coupling (PAC)**, where the phase of hippocampal theta oscillations modulates the amplitude of gamma activity in downstream regions. We expect this modulation to be time-locked to the cognitive processing of the stimulus (approximately 300–600 ms post-stimulus), reflecting the transmission of prediction error signals required for model updating.

II. Methods

A. Feature Extraction: Phase-Amplitude Coupling (PAC)

To elucidate the temporal dynamics of information flow during novelty processing, we transitioned from raw Local Field Potential (LFP) signals to a feature space based on functional connectivity. Specifically, we calculated Phase-Amplitude Coupling (PAC) to quantify how the phase of low-frequency oscillations (Theta: 4–8 Hz) modulates the amplitude of high-frequency activity (Gamma: 30–80 Hz) across brain regions. This measure is particularly relevant for studying cross-regional communication during learning and memory tasks [10], [11].

The continuous LFP data were segmented into trials based on event markers (Standard vs. Target). To capture the time course of neural responses, we analyzed the data in non-overlapping sliding windows of 200 ms. The choice of 200 ms provides a balance between temporal resolution (to detect transient P300-like events) and frequency resolution (sufficient cycles for Theta estimation). For each window, we computed the Mean Vector Length (MVL) as a metric of coupling strength via the Hilbert transform. The resulting feature space had dimensions of $Sessions \times RegionPairs \times Trials \times TimeWindows$.

B. Directional Causality

To further investigate the directionality of information flow and test our specific hypothesis regarding hippocampal dominance, we employed Granger Causality (GC) analysis. Unlike PAC, which establishes coupling but not necessarily direction, GC tests whether the past values of a source region (Hippocampus) can predict the future values of a target region (Dorsal Striatum) better than the target's own past values alone. We computed the F-statistic for the HC $\rightarrow$ dS pathway specifically in the time window of peak activity (Window 5).

C. Statistical Analysis

Hypothesis testing was performed to compare the distribution of connectivity features between Standard and Target (novel) conditions. Given the unequal number of trials ($N_{std} \gg N_{tgt}$) and the potential for unequal variances, we employed Welch's t-test for independent samples.

Since our hypothesis explicitly predicted an increase in directional coupling for novel stimuli, we utilized a one-sided (right-tailed) Welch's t-test. The null hypothesis (H_0) posited that the mean information flow is equal between or lower for target trials compared to standard trials. We performed this test separately for each time window with a significance threshold of $\alpha = 0.05$.

Furthermore, because our analysis involved statistical tests across multiple non-overlapping 200 ms time windows, there is an inherent risk of inflating the Type I error rate due to multiple comparisons. While we report the uncorrected p-values to capture the exploratory temporal dynamics of this pathway, we acknowledge this limitation; however, since our findings yielded non-significant results ($p > 0.05$), applying a strict correction (e.g., False Discovery Rate or Bonferroni) would only make the thresholds more conservative, thereby reinforcing our conclusion to retain the null hypothesis. All analyses were implemented in Python using SciPy and Statsmodels libraries.

III. Results

A. Temporal Dynamics of Hippocampal-Striatal Connectivity

We investigated whether the presentation of novel stimuli modulates the directional information flow between the Hippocampus (HC) and Dorsal Striatum (dS). Our analysis focused on the temporal evolution of Theta-Gamma Phase-Amplitude Coupling (PAC) across 200 ms time windows following stimulus onset.

1) Robust Temporal Modulation: As illustrated in Figure 1 (Left panel), the network exhibited a striking temporal response. We observed a robust, time-dependent increase in PAC strength starting from Window 4 (approx. 130–330 ms) and peaking in Windows 5 and 6 (330–730 ms). This indicates that the hippocampal-striatal circuit is strongly recruited during the late latency processing of auditory stimuli, coinciding with the cognitive processing timeframes (e.g., P300).

2) Hypothesis Testing (Standard vs. Target): We tested the primary hypothesis that this coupling would be stronger for novel (Target) stimuli. Contrary to prediction, statistical analysis using Welch's t-test revealed no significant difference between Standard and Target conditions in the HC-dS pathway across the analyzed time windows. As seen in the figure, the trajectories for both conditions overlap almost perfectly. For the peak activity window (Win 5), the p-value was 0.49 ($p > 0.05$), leading us to retain the null hypothesis.

B. Testing Hypothesis

Hypothesis: We formulated an additional hypothesis concerning directionality: *"If the hippocampus acts as a novelty detector, the directional drive from HC to dS (Granger Causality) should be significantly higher for Target trials during the cognitive evaluation window (Window 5)."*

Result: To test this, we computed the Granger Causality F-score for the HC $\rightarrow$ dS connection. As shown in Figure 1 (Right panel), while the causality scores were substantial (indicating information flow exists), the comparison between Standard ($Mean \approx 5.2$) and Target ($Mean \approx 5.7$) trials did not yield a statistically significant difference ($t = 0.82$, $p = 0.435$).

This result converges with our PAC findings, suggesting that while the HC $\rightarrow$ dS pathway is active during the task, its engagement intensity is likely driven by the general processing of auditory sequences rather than being a specific filter for stimulus rarity in this dataset.

IV. Discussion

The primary goal of this study was to identify network-level signatures of novelty processing by moving beyond local neural responses to dynamic functional connectivity. We hypothesized that the detection of novel stimuli is mediated by a transient increase in directional information flow from the Hippocampus to the Dorsal Striatum (HC-dS) and Medial Prefrontal Cortex (mPFC). Here, we interpret our findings in the context of existing theories of surprise and learning.

A. Evaluation of Hypotheses

Our central hypothesis posited that novelty (Target stimuli) would elicit significantly stronger Theta-Gamma Phase-Amplitude Coupling (PAC) and directional causality compared to standard stimuli. **This hypothesis was not supported by the statistical results.** We found no significant difference in coupling strength between conditions ($p > 0.05$) in either the HC-mPFC or HC-dS pathways.

However, our analysis revealed a secondary, crucial finding: a **robust, time-locked modulation** of the HC-dS circuit. Regardless of stimulus type, the coupling strength surged specifically in the 330–730 ms latency range. This suggests that while the hippocampal-striatal loop is strongly engaged during late-stage auditory processing, its Theta-Gamma synchronization serves a more general role in orienting or temporal integration rather than acting as a selective filter for stimulus probability in this paradigm.

B. Integration with Prior Literature

The timing of the observed connectivity modulation (peaking around 300–500 ms) aligns remarkably well with the temporal window of the **P300 (P3b)** component, which is traditionally linked to context updating and memory consolidation [8]. Previous ERP studies have consistently shown that P300 amplitude is larger for deviant targets [7]. Our LFP results offer a nuanced perspective: while the *amplitude* of local activity (ERP) might discriminate novelty, the *temporal coordination* (PAC) between Hippocampus and Striatum appears to be engaged by the stimulus stream generally.

This finding partially challenges the view that all network interactions in this window must be novelty-specific. Instead, it supports models where the Hippocampus acts as a continuous comparator [11], constantly evaluating incoming inputs (both standard and target) and maintaining a baseline level of communication with the striatum to regulate potential gating, which may only differentiate in magnitude under active behavioral demands.

C. Novelty and Theoretical Contribution

This study distinguishes itself from prior work by shifting the focus from **local activation** to **dynamic information flow**. Most traditional studies rely on static snapshots of connectivity or averaged ERPs [6]. In contrast, our approach utilized time-resolved PAC and Granger Causality on LFP data, allowing us to map the precise temporal evolution of circuit interactions.

A key contribution of this work is the demonstration that the HC-dS pathway exhibits specific temporal dynamics (a late-latency surge) that are distinct from the early sensory processing phase. This validates the feasibility of using LFP features to track real-time network reorganization, providing a methodological framework that extends beyond component-based analysis.

D. Limitations and Future Directions

Our study has limitations that frame the interpretation of the null results.

Methodological Limitation: We focused exclusively on Theta-Gamma coupling. It is possible that the novelty signal is multiplexed in other frequency bands, such as Beta oscillations (15-30 Hz), which are increasingly implicated in maintaining current cognitive sets and signaling prediction errors [10].

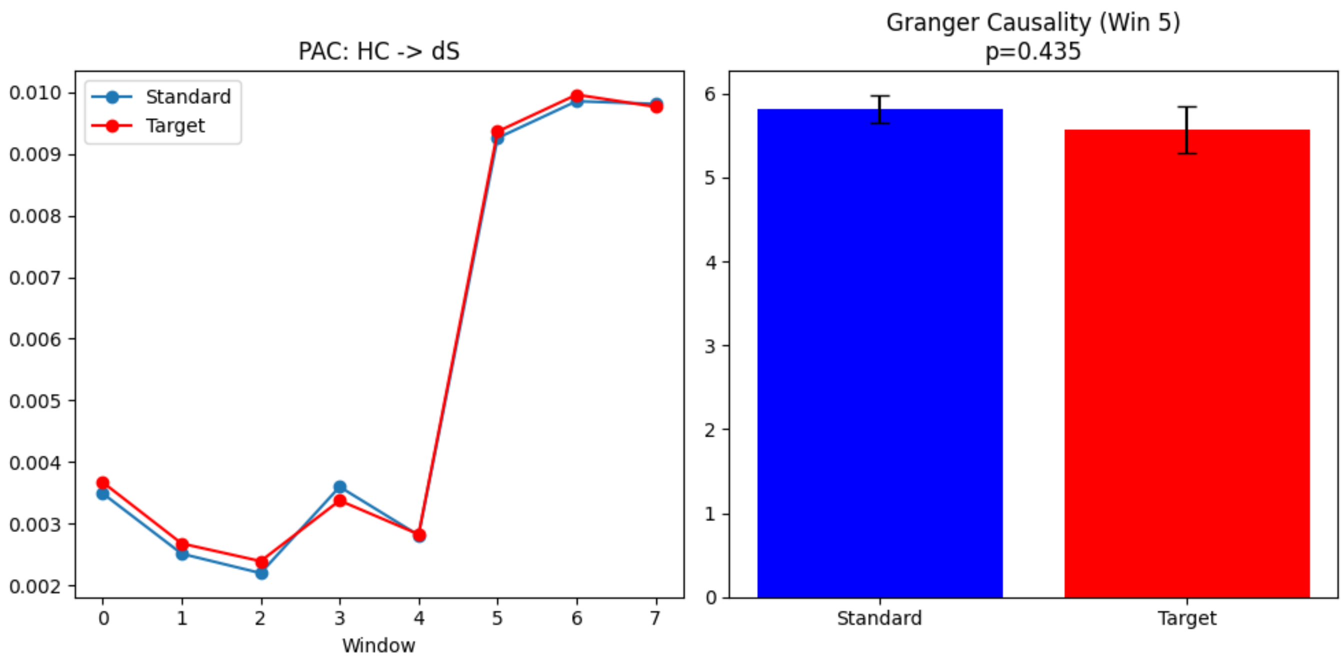

Fig. 1. **Analysis of Hippocampal-Striatal Connectivity. Left:** Temporal dynamics of Phase-Amplitude Coupling (PAC) for the HC-dS pathway. The plot shows the mean Coupling Strength (MVL) across time windows. Note the robust increase in coupling strength starting from Window 4 (approx. 130 ms) and peaking in Windows 5-6 for *both* Standard (blue) and Target (red) conditions. **Right:** Testing the directional hypothesis via Granger Causality (HC $\rightarrow$ dS) in the peak window (Win 5). Error bars represent SEM. No statistically significant difference was observed between conditions ($p = 0.435$).

Experimental Limitation: The dataset was derived from a passive auditory oddball task. Without a behavioral requirement (e.g., button press), the "significance" of the target stimuli is reduced, which might attenuate the top-down modulation from mPFC and the dopaminergic drive to the striatum [13].

Future Directions: Future research should employ active discrimination tasks to enhance the salience of deviant stimuli. Additionally, employing cross-frequency coupling measures across a broader spectral range (e.g., Delta-Beta) could reveal frequency-specific channels of novelty transmission that were not captured in this study.

Acknowledgments and AI Declaration

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